

Basic Electronics All MCQ's

- limited edition

UNIT-1

Question 1

If a 2 mV input signal produces a 2V output signal, what is the voltage gain?

Select one:

- ☐ a. 0.004
- ☐ b. 1000
- ☐ c. 100
- ☐ d. 0.001

Feedback

The correct answer is: 1000

Question 2

For a JFET, when gate to source voltage is made sufficiently negative, the drain current reaches zero mA when

Select one:

- ☐ a. $V_{GS} = V_P$
- ☐ b. $V_{GS} = V_P$
- ☐ c. $V_{GS} = -V_P$
- ☐ d. $V_{GS} = I_D$

Feedback

The correct answer is: $V_{GS} = -V_P$

Question 3

When BJT operates as a switch, it is operated alternately in _____

Select one:

- ☐ a. Saturation and active region
- ☐ b. Active and saturation region
- ☐ c. Cut-off and Saturation region
- ☐ d. Active and cut-off region

Feedback

The correct answer is: Cut-off and Saturation region

Question 4

JFET is considered as a voltage-controlled device because _____. ..

Select one:

- ☐ a. Gate current is controlled by drain voltage
- ☐ b. Drain current is controlled by gate voltage

- ☐ c. Drain current is controlled by source voltage
- ☐ d. Gate current is controlled by source voltage

Feedback

The correct answer is: Drain current is controlled by gate voltage

Question 5

JFET acts as constant current source above

Select one:

- ☐ a. Threshold voltage
- ☐ b. Breakdown voltage
- ☐ c. None
- ☐ d. Pinch off voltage

Feedback

The correct answer is: Pinch off voltage

Question 6

The efficiency of full wave rectifier is about

Select one:

- ☐ a. 40.6
- ☐ b. 1.21
- ☐ c. 81.2
- ☐ d. 0.43

Feedback

The correct answer is: 81.2

Question 7

When forward biased, a p-n junction diode

Select one:

- ☐ a. Conducts current
- ☐ b. has high resistance
- ☐ c. low voltage
- ☐ d. Blocks current

Feedback

The correct answer is: Conducts current

Question 8

In CE-RC coupled amplifier, an emitter bypass capacitor is connected in parallel with emitter resistance R_E to

Select one:

- ☐ a. To support the resistance R_E

- ☐ b. Increase the noise in the circuit
- ☐ c. To provide a low reactance path to the amplified ac signal
- ☐ d. To support the resistance R_E

Feedback

The correct answer is: To provide a low reactance path to the amplified ac signal

Question 9

Zener diode is mainly used as

Select one:

- ☐ a. Oscillator
- ☐ b. Comparator
- ☐ c. Amplifier
- ☐ d. Voltage regulator

Feedback

The correct answer is: Voltage regulator

Question 10

A transistor has $I_B = 0.06$ mA and $I_E = 9.00$ mA. The value of β is

Select one:

- ☐ a. 99
- ☐ b. 149
- ☐ c. 119
- ☐ d. 129

Feedback

The correct answer is: 149

Question 11

Calculate I_D in n- JFET, Given $V_{GS} = -2V$, $V_P = -4V$ and $I_{DSS} = 4mA$

Select one:

- ☐ a. 1mA
- ☐ b. 3mA
- ☐ c. 4mA
- ☐ d. 2mA

Feedback

The correct answer is: 1mA

Question 12

. When a JFET is cut off, it is like a switch and when it is saturated, it is like a switch

Select one:

- ☐ a. Closed, closed
- ☐ b. Closed, open
- ☐ c. Open, closed
- ☐ d. Open, open

Feedback

The correct answer is: Open, closed

Question 13

In CE configuration output VI characteristics is plotted by taking_____

Select one:

- ☐ a. V_{CE} verses I_C for constant I_B
- ☐ b. V_{CE} verses I_C for constant V_{CB}
- ☐ c. V_{CE} verses I_C for constant I_E
- ☐ d. V_{CE} verse I_E for constant I_C

Feedback

The correct answer is: V_{CE} verses I_C for constant I_B

Question 14

The value of α for a silicon transistor, given the collector current as 1 mA and base current as 25 μ A is

Select one:

- ☐ a. 0.9376
- ☐ b. 0.9756
- ☐ c. 0.9176
- ☐ d. 0.9276

Feedback

The correct answer is: 0.9756

Question 15

In a CE amplifier circuit, the output signal generates a phase shift of

Select one:

- ☐ a. 180°
- ☐ b. 270°
- ☐ c. 0°
- ☐ d. 90°

Feedback

The correct answer is: 180°

Question 16

The 7805 voltage regulator IC produces output voltage of

Select one:

- ☐ a. $\pm 5 \text{ V}$
- ☐ b. 78 V
- ☐ c. 10V
- ☐ d. 5 V

Feedback

The correct answer is: 5 V

Question 17

Average DC load voltage of a full wave Bridge rectifier is

Select one:

- ☐ a. $2I_m/\pi$
- ☐ b. $I_m/0.5$
- ☐ c. $2V_m/\pi$
- ☐ d. $I_m/0.707$

Feedback

The correct answer is: $2V_m/\pi$

Question 18

In the breakdown region, zener diode behaves like a source of

Select one:

- ☐ a. Constant power
- ☐ b. Constant current
- ☐ c. Constant resistance
- ☐ d. Constant voltage

Feedback

The correct answer is: Constant voltage

Question 19

The MOSFET stands for _____

Select one:

- ☐ a. Metal oxide semiconductor FET
- ☐ b. Metal oxidized selenium FET
- ☐ c. Metal oxide surface FET
- ☐ d. Metal of surface FET

Feedback

The correct answer is: Metal oxide semiconductor FET

Question 20

In n- JFET the drain current is 2mA , when $V_{GS} = -8\text{V}$ and $V_P = -8\text{V}$, the values of I_{DSS} is

Select one:

- ☐ a. 1mA
- ☐ b. 4mA
- ☐ c. 2mA
- ☐ d. 3mA

Feedback

The correct answer is: 2mA

Question 21

Not answered
Marked out of 1.00

Flag question

Question text

Which transistor element is used in CMOS logic?

Select one:

- ☐ a. Unijunction
- ☐ b. JFET
- ☐ c. MOSFET
- ☐ d. Bipolar

Feedback

The correct answer is: MOSFET

Question 22

Not answered
Marked out of 1.00

Flag question

Question text

Which is not a MOSFET terminal?

Select one:

- ☐ a. Drain
- ☐ b. Gate
- ☐ c. Source
- ☐ d. Base

Feedback

The correct answer is: Base

Question 23

Not answered

Marked out of 1.00

Flag question

Question text

When the p-n junction diode is forward biased, it offers

Select one:

- ☐ a. high voltage
- ☐ b. high resistance
- ☐ c. low voltage
- ☐ d. low resistance

Feedback

The correct answer is: low resistance

Question 24

Not answered

Marked out of 1.00

Flag question

Question text

Amplifiers and oscillators using BJT are operated in region

Select one:

- ☐ a. Saturation
- ☐ b. Breakdown
- ☐ c. Cut-off
- ☐ d. Active

Feedback

The correct answer is: Active

Question 25

What is the resistance of an ideal diode in the region of conduction?

Select one:

- ☐ a. $0\ \Omega$
- ☐ b. $5\ \text{k}\Omega$
- ☐ c. Unpredictable
- ☐ d. Infinity

Feedback

The correct answer is: $0\ \Omega$

Question 26

Not answered
Marked out of 1.00

Flag question

Question text

When the p-n junction diode is reverse biased, it offers

Select one:

- ☐ a. low voltage
- ☐ b. high voltage
- ☐ c. low resistance
- ☐ d. high resistance

Feedback

The correct answer is: high resistance

Question 27

The signal voltage gain of an amplifier, A_v , is defined as _____.

Select one:

- ☐ a. $A_v = I_c * R_c$
- ☐ b. $A_v = V_{in}/V_{out}$
- ☐ c. $A_v = V_{out}/V_{in}$
- ☐ d. $A_v = V_{BE}/V_{CC}$

Feedback

The correct answer is: $A_v = V_{out}/V_{in}$

Question 28

Not answered
Marked out of 1.00

Flag question

Question text

In the ohmic region, the JFET can be used as

Select one:

- ☐ a. Voltage variable current
- ☐ b. Voltage variable capacitor
- ☐ c. Voltage variable inductor
- ☐ d. Voltage variable resistor

Feedback

The correct answer is: Voltage variable resistor

Question 29

Not answered
Marked out of 1.00

Flag question

Question text

In a bipolar junction transistor the collector current is controlled by

Select one:

- ☐ a. Collector resistance
- ☐ b. None
- ☐ c. Collector voltage
- ☐ d. Base current

Feedback

The correct answer is: Base current

Question 30

Not answered
Marked out of 1.00

Flag question

Question text

The value of β for a silicon transistor given the collector current as 1 mA and base current as 25 μ A is

Select one:

- ☐ a. 40
- ☐ b. 4
- ☐ c. 4000
- ☐ d. 400

Feedback

The correct answer is: 40

Question 31

Not answered

Marked out of 1.00

Flag question

Question text

The leakage current in a transistor is due to:

Select one:

- ☐ a. Majority charge carriers
- ☐ b. Minority charge carriers
- ☐ c. Zener effect
- ☐ d. Breakdown

Feedback

The correct answer is: Minority charge carriers

Question 32

Not answered

Marked out of 1.00

Flag question

Question text

The amount of ripples present at the output of half wave rectifier is

Select one:

- ☐ a. 0.46
- ☐ b. 0.5
- ☐ c. 1.21
- ☐ d. 0.81

Feedback

The correct answer is: 1.21

Question 33

Not answered

Marked out of 1.00

Flag question

Question text

The charge carriers in an N-channel JFET are

Select one:

- ☐ a. Protons
- ☐ b. Neutrons
- ☐ c. Electrons
- ☐ d. Holes

Feedback

The correct answer is: Electrons

Question 34

Not answered

Marked out of 1.00

Flag question

Question text

A germanium diode is connected in series to a dc supply of 7.5 volts and resistance of 1.3 k Ω , the current in the circuit is

Select one:

- ☐ a. 5.53 A
- ☐ b. 1.53 mA
- ☐ c. 5.53 mA

☐ d. 5.35 mA

Feedback

The correct answer is: 5.53 mA

Question 35

Not answered

Marked out of 1.00

Flag question

Question text

A transistor has $I_B = 0.08$ mA and $I_E = 9.60$ mA. The value of β is

Select one:

- ☐ a. 139
- ☐ b. 119
- ☐ c. 99
- ☐ d. 129

Feedback

The correct answer is: 119

Question 36

Not answered

Marked out of 1.00

Flag question

Question text

For a BJT, the following statement is true

Select one:

- ☐ a. $I_B = I_E$
- ☐ b. $I_E = I_C$
- ☐ c. $I_B = \alpha I_E$
- ☐ d. $I_B = I_C$

Feedback

The correct answer is: $I_E = I_C$

Question 37

Not answered

Marked out of 1.00

Flag question

Question text

The value of β for a silicon transistor given the collector current as 1 mA and base current as 10 μ A is

Select one:

- ☐ a. 0.1
- ☐ b. 1000
- ☐ c. 10
- ☐ d. 100

Feedback

The correct answer is: 100

Question 38

Not answered
Marked out of 1.00

Flag question

Question text

A germanium diode is connected in series to a dc supply of 4V and resistance of 2 k Ω , the current in the circuit is

Select one:

- ☐ a. 1.85 mA
- ☐ b. 0.185 mA
- ☐ c. 18.5 mA
- ☐ d. 1.45mA

Feedback

The correct answer is: 1.85 mA

Question 39

Not answered
Marked out of 1.00

Flag question

Question text

MOSFET has greatest application in digital circuit due to

Select one:

- ☐ a. Small amount of space it takes on a chip
- ☐ b. None
- ☐ c. Less noise
- ☐ d. Low power consumption

Feedback

The correct answer is: Low power consumption

Question 40

Not answered
Marked out of 1.00

Flag question

Question text

Higher the ripple factor, the output will have higher components of

Select one:

- ☐ a. pulse
- ☐ b. DC
- ☐ c. AC
- ☐ d. spike

Feedback

The correct answer is: AC

Question 41

Not answered
Marked out of 1.00

Flag question

Question text

A transistor has $I_B = 0.06 \text{ mA}$ and $I_E = 9.00 \text{ mA}$. The value of α is

Select one:

- ☐ a. 990
- ☐ b. 9.9
- ☐ c. 0.099
- ☐ d. 0.99

Feedback

The correct answer is: 0.99

Question 42

Not answered

Marked out of 1.00

Flag question

Question text

Zener diode regulates output voltage only when it is connected in

Select one:

- ☐ a. Neither in forward or reverse bias
- ☐ b. forward bias only
- ☐ c. reverse bias only
- ☐ d. both in forward and reverse bias

Feedback

The correct answer is: reverse bias only

Question 43

Not answered

Marked out of 1.00

Flag question

Question text

The r.m.s value of load current in a half wave rectifier is

Select one:

- ☐ a. $0.5 I_m$
- ☐ b. $0.707 I_m$

- ☐ c. $1.414 I_m$
- ☐ d. $1.212 I_m$

Feedback

The correct answer is: $0.5 I_m$

Question 44

Not answered

Marked out of 1.00

Flag question

Question text

The voltage at which forward current through the diode starts increasing rapidly is called as

Select one:

- ☐ a. Cut off voltage
- ☐ b. Saturation voltage
- ☐ c. Cut in voltage
- ☐ d. Breakdown voltage

Feedback

The correct answer is: Cut in voltage

Question 45

Not answered

Marked out of 1.00

Flag question

Question text

Base terminal of the transistor in Common Base configuration is connected to

Select one:

- ☐ a. Kept open
- ☐ b. Input and output
- ☐ c. Output only
- ☐ d. Input only

Feedback

The correct answer is: Input and output

Question 46

Not answered

Marked out of 1.00

Flag question

Question text

The amount of ripples present at the output of a full wave rectifier is

Select one:

- ☐ a. 81%
- ☐ b. 48%
- ☐ c. 92%
- ☐ d. 54%

Feedback

The correct answer is: 48%

Question 47

Not answered
Marked out of 1.00

Flag question

Question text

The range of collector to emitter current gain α is

Select one:

- ☐ a. 20 to 200
- ☐ b. 0.76 to 0.96
- ☐ c. 0.96 to 0.99
- ☐ d. 0.50 to 1

Feedback

The correct answer is: 0.96 to 0.99

Question 48

Not answered
Marked out of 1.00

Flag question

Question text

A highly stable biasing network used in CE-RC coupled amplifier is _____

Select one:

- ☐ a. Emitter bias circuit
- ☐ b. Voltage divider biasing circuit
- ☐ c. Base bias circuit
- ☐ d. Collector feedback bias circuit

Feedback

The correct answer is: Voltage divider biasing circuit

Question 49

Not answered
Marked out of 1.00

Flag question

Question text

When n-channel JFET with gate terminal open circuited and a positive voltage V_{DS} is applied across drain and source terminals, depletion region is wider near

Select one:

- ☐ a. None
- ☐ b. Source
- ☐ c. Gate
- ☐ d. Drain

Feedback

The correct answer is: Drain

Question 50

Not answered
Marked out of 1.00

Flag question

Question text

A transistor has a beta of 200 and a base current of $40\text{ }\mu\text{A}$ in CE configuration. Then the collector current is

Select one:

- ☐ a. 8 mA
- ☐ b. $8\mu\text{A}$
- ☐ c. 80 mA
- ☐ d. $80\text{ }\mu\text{A}$

Feedback

The correct answer is: 8 mA

Question 51

Not answered
Marked out of 1.00

Flag question

Question text

A sinusoidal voltage of peak value 40V is applied to a half wave rectifier with $R_L = 800\text{ }\Omega$ and $R_f = 8\text{ }\Omega$. The peak value of the current is _____

Select one:

- ☐ a. 4.95 mA
- ☐ b. 4.95 A
- ☐ c. 0.495 mA
- ☐ d. 49.5 mA

Feedback

The correct answer is: 49.5 mA

Question 52

Not answered
Marked out of 1.00

Flag question

Question text

n-channel FETs are superior to p-channel FETs because

Select one:

- ☐ a. Mobility of electrons is greater than that of holes
- ☐ b. Mobility of electrons is smaller than that of holes
- ☐ c. They have high switching time
- ☐ d. They consume less power

Feedback

The correct answer is: Mobility of electrons is greater than that of holes

Question 53

Not answered

Marked out of 1.00

Flag question

The collector to base current gain β in terms of α is defined as

Select one:

- ☐ a. $\frac{\alpha}{1+\alpha}$
- ☐ b. $\frac{\alpha}{1-\alpha}$
- ☐ c. $\frac{\beta}{1+\alpha}$
- ☐ d. $\frac{\beta}{1+\beta}$

The correct answer is: $\frac{\alpha}{1-\alpha}$

Question 54

Not answered

Marked out of 1.00

Flag question

Question text

Which of the following electrical characteristics is not exhibited by an ideal op-amp?

Select one:

- ☐ a. Infinite voltage gain
- ☐ b. Infinite bandwidth
- ☐ c. Infinite output resistance
- ☐ d. Infinite slew rate

Feedback

The correct answer is: Infinite output resistance

Question 55

Not answered

Marked out of 1.00

Flag question

Question text

A resistance is connected in series with zener diode in a voltage regulator to

Select one:

- ☐ a. switch off the zener
- ☐ b. Properly forward bias the zener
- ☐ c. Properly reverse bias the zener
- ☐ d. Protect the zener

Feedback

The correct answer is: Protect the zener

Question 56

Not answered

Marked out of 1.00

Flag question

Question text

A JFET is a.....controlled device whereas a bipolar transistor is a..... controlled device.

Select one:

- ☐ a. current, current

- ☐ b. voltage, current
- ☐ c. current, voltage
- ☐ d. None of these

Feedback

The correct answer is: voltage, current

Question 57

Not answered
Marked out of 1.00

Flag question

Question text

When the transistor operates as an amplifier, the operating point is located_____

Select one:

- ☐ a. In the Ohmic region
- ☐ b. In the center of active region
- ☐ c. In the cut-off region
- ☐ d. In the Saturation region

Feedback

The correct answer is: In the center of active region

Question 58

Not answered
Marked out of 1.00

Flag question

Question text

A Zener diode is a heavily doped semiconductor device that is designed to operate in

Select one:

- ☐ a. Forward direction
- ☐ b. None
- ☐ c. Both forward and reverse
- ☐ d. Reverse direction

Feedback

The correct answer is: Reverse direction

Question 59

Not answered

Marked out of 1.00

[Flag question](#)

Question text

The r.m.s value of load current in a full wave rectifier is

Select one:

- ☐ a. $0.707 I_m$
- ☐ b. 0.5π
- ☐ c. $0.5 I_m$
- ☐ d. $0.3183 I_m$

Feedback

The correct answer is: $0.707 I_m$

Question 60

Not answered

Marked out of 1.00

[Flag question](#)

Question text

A Zener diode of 10V is connected to a supply of 20V through a series resistance of 200 ohms. With no load, the current through the Zener is

Select one:

- ☐ a. 50 mA
- ☐ b. 400 mA
- ☐ c. 5 A
- ☐ d. 500 mA

Feedback

The correct answer is: 50 mA

Question 61

Not answered

Marked out of 1.00

Flag question

Question text

The range of collector to base current gain β is

Select one:

- ☐ a. 20 to 100
- ☐ b. 20 to 200
- ☐ c. 50 to 100
- ☐ d. 10 to 50

Feedback

The correct answer is: 20 to 200

Question 62

Not answered
Marked out of 1.00

Flag question

Question text

A silicon and a germanium diodes are connected in series to a dc supply of 8 volts and resistance of $2.5 \text{ k}\Omega$, the current in the circuit is

Select one:

- ☐ a. 28mA
- ☐ b. 0.28mA
- ☐ c. 2.8mA
- ☐ d. 2.8A

Feedback

The correct answer is: 2.8mA

Question 63

Not answered
Marked out of 1.00

Flag question

Question text

A bipolar junction transistor is a

Select one:

- ☐ a. Current controlled device
- ☐ b. Resistance controlled device
- ☐ c. Junction controlled device
- ☐ d. Voltage controlled device

Feedback

The correct answer is: Current controlled device

Question 64

Not answered

Marked out of 1.00

 Flag question

The collector to emitter current gain α in terms of β is defined as

Select one:

- ☐ a. $\frac{\alpha}{1+\alpha}$
- ☐ b. $\frac{\beta}{1+\alpha}$
- ☐ c. $\frac{\beta}{1+\beta}$
- ☐ d. $\frac{\alpha}{1-\alpha}$

The correct answer is: $\frac{\beta}{1+\beta}$

Question 65

Not answered

Marked out of 1.00

 Flag question

Question text

A transistor has $I_B = 0.08$ mA and $I_E = 9.60$ mA. The value of α is

Select one:

- ☐ a. 990
- ☐ b. 100
- ☐ c. 0.99
- ☐ d. 9.9

Feedback

The correct answer is: 0.99

Question 66

Not answered

Marked out of 1.00

Flag question

Question text

Higher the ripple factor, the output will have smaller components of

Select one:

- ☐ a. AC
- ☐ b. DC
- ☐ c. Spike
- ☐ d. Pulse

Feedback

The correct answer is: DC

Question 67

Not answered

Marked out of 1.00

Flag question

Question text

The total emitter current in BJT is

Select one:

- ☐ a. $I_C + I_E$
- ☐ b. $I_B + I_C$
- ☐ c. $I_C + I_{CBO}$
- ☐ d. $I_B - I_C$

Feedback

The correct answer is: $I_B + I_C$

Question 68

Not answered

Marked out of 1.00

Flag question

Question text

For a JFET, above the pinch-off voltage, the drain current

Select one:

- ☐ a. Increases
- ☐ b. Remains constant
- ☐ c. Decreases
- ☐ d. Varies parabolically

Feedback

The correct answer is: Remains constant

Question 69

Not answered

Marked out of 1.00

Flag question

Question text

The value of α for a silicon transistor, given the collector current as 1 mA and base current as 10 μ A is

Select one:

- ☐ a. 0.9376
- ☐ b. 0.9900
- ☐ c. 0.9276
- ☐ d. 0.9176

Feedback

The correct answer is: 0.9900

Question 70

Not answered

Marked out of 1.00

Flag question

Question text

In JFET, the drain voltage above which there is no increase in the drain current is called _____ voltage

Select one:

- ☐ a. Critical voltage
- ☐ b. Breakdown voltage
- ☐ c. Pinch off voltage

Feedback

The correct answer is: Pinch off voltage

Question 71

Not answered
Marked out of 1.00

Flag question

Question text

The DC load current of a half wave rectifier is _____

Select one:

- ☐ a. $1.414/\pi$
- ☐ b. I_m/π
- ☐ c. $2I_{dc}/\pi$
- ☐ d. $2I_m/\pi$

Feedback

The correct answer is: I_m/π

Question 72

Not answered
Marked out of 1.00

Flag question

Question text

What is the state of an ideal diode in the region of non-conduction?

Select one:

- ☐ a. Short circuit
- ☐ b. Unpredictable
- ☐ c. Undefined
- ☐ d. Open circuit

Feedback

The correct answer is: Open circuit

Question 73

Not answered
Marked out of 1.00

Flag question

Question text

The enhancement type MOSFET works only with _____.

Select one:

- ☐ a. large negative drain voltage
- ☐ b. large positive gate voltage
- ☐ c. large negative gate voltage
- ☐ d. large positive drain voltage

Feedback

The correct answer is: large positive gate voltage

Question 74

Not answered
Marked out of 1.00

Flag question

Question text

The efficiency of full wave rectifier is about

Select one:

- ☐ a. 1.21%
- ☐ b. 40.6 %
- ☐ c. 81.2%

☐ d. 46 %

Feedback

The correct answer is: 81.2%

Question 75

Not answered

Marked out of 1.00

[Flag question](#)

Question text

Smaller the ripple factor, the output will have higher components of

Select one:

- ☐ a. DC
- ☐ b. AC
- ☐ c. Spike
- ☐ d. Pulse

Feedback

The correct answer is: DC

Question 76

Not answered

Marked out of 1.00

[Flag question](#)

Question text

A silicon diode is connected in series to a dc supply of 5 Volts and resistance of 1.5 k Ω , the current in the circuit is

Select one:

- ☐ a. 2.86 mA
- ☐ b. 1.5mA
- ☐ c. 3.86 mA
- ☐ d. 0.38 mA

Feedback

The correct answer is: 2.86 mA

Question 77

Not answered
Marked out of 1.00

Flag question

Question text

A transistor has a beta of 200 and a base current of $40\text{ }\mu\text{A}$ in CE configuration. Then the emitter current is

Select one:

- ☐ a. $80.04\text{ }\mu\text{A}$
- ☐ b. 8.04mA
- ☐ c. 80 mA
- ☐ d. $80\mu\text{A}$

Feedback

The correct answer is: 8.04mA

Question 78

Not answered
Marked out of 1.00

Flag question

Question text

A silicon diode is connected in series to a dc supply of 2 volts and resistance of $13\text{ }\Omega$, the current in the circuit is_____

Select one:

- ☐ a. 0.10A
- ☐ b. 1.5A
- ☐ c. 0.45 A
- ☐ d. 0.55 A

Feedback

The correct answer is: 0.10A

Question 79

Not answered
Marked out of 1.00

Flag question

Question text

For a JFET, as V_{GS} is made more and more negative, saturation level of I_D reduces further and the pinch-off voltage continues to drop in

Select one:

- ☐ a. Parabolic form
- ☐ b. Exponential form
- ☐ c. Linear form
- ☐ d. None

Feedback

The correct answer is: Parabolic form

Question 80

Not answered
Marked out of 1.00

Flag question

Question text

Whenever a JFET operates above pinch-off voltage

Select one:

- ☐ a. Drain current increases steeply
- ☐ b. Drain current starts decreasing
- ☐ c. Depletion regions become smaller
- ☐ d. Drain current remains nearly constant

The correct answer is: Drain current remains nearly constant

UNIT-2

Question 1

Not answered
Marked out of 1.00

Flag question

Question text

An RC network in an RC phase shift oscillator has $C = 0.1 \mu\text{F}$. The frequency of oscillations is 1 kHz. The value of R used in the feedback network is

Select one:

- ☐ a. 600Ω
- ☐ b. $1 \text{ k}\Omega$
- ☐ c. 500Ω
- ☐ d. 650Ω

Feedback

The correct answer is: 650Ω

Question 2

Not answered
Marked out of 1.00

Flag question

Question text

An ideal OP-AMP has following characteristics

Select one:

- ☐ a. An ideal OP-AMP has following characteristics
- ☐ b. $R_i = 0$, $A = \text{infinity}$, $R_o = \text{infinity}$
- ☐ c. $R_i = \text{infinity}$, $A = 0$, $R_o = \text{infinity}$
- ☐ d. $R_i = \text{infinity}$, $A = \text{infinity}$, $R_o = \text{infinity}$
- ☐ e. $R_i = 0$, $A = \text{infinity}$, $R_o = \text{zero}$

Feedback

The correct answer is: An ideal OP-AMP has following characteristics

Question 3

Not answered
Marked out of 1.00

Flag question

Question text

The output voltage of an Op-amp integrator is given by

Select one:

- ☐ a. $V_o = -1/(R_1 C_f) \int_0^t V_{in} dt$
- ☐ b. $V_o = -1/(R_1 C_f) \int_{-\infty}^t V_{in} dt$
- ☐ c. $V_o = -1/(C_f) \int_0^t V_{in} dt$
- ☐ d. $V_o = -1/(R_f C_1) \int_0^t V_{in} dt$

Feedback

The correct answer is: $V_o = -1/(R_1 C_f) \int_0^t V_{in} dt$

Question 4

Not answered
Marked out of 1.00

Flag question

Question text

Find the output voltage of an ideal op-amp if V_1 and V_2 are the two input voltages

Select one:

- ☐ a. $V_o = V_1 * V_2$
- ☐ b. $V_o = A(V_1 - V_2)$
- ☐ c. $V_o = V_1 - V_2$
- ☐ d. $V_o = A(V_1 + V_2)$

Feedback

The correct answer is: $V_o = A(V_1 - V_2)$

Question 5

Not answered
Marked out of 1.00

Flag question

Question text

The outputs of two comparators are connected to a in the internal circuitry of IC 555 timer

Select one:

- ☐ a. D-type Flip Flop
- ☐ b. S-R Flip Flop
- ☐ c. T-Type Flip Flop
- ☐ d. J-K Flip Flop

Feedback

The correct answer is: S-R Flip Flop

Question 6

Not answered
Marked out of 1.00

Flag question

Question text

A sine wave of 0.5 V peak voltage is applied as an input to an inverting amplifier with $R_1 = 10 \text{ k}\Omega$ and $R_f = 50 \text{ k}\Omega$. The output voltage V_o is

Select one:

- ☐ a. -10 V peak
- ☐ b. 5.2 V peak
- ☐ c. -2.5 V peak
- ☐ d. -2.5 V peak-peak

Feedback

The correct answer is: -2.5 V peak

Question 7

Not answered
Marked out of 1.00

Flag question

Question text

The identification 555 for IC 555 timer is mainly because

Select one:

- ☐ a. It has five Op Amp comparators internally

- ☐ b. It has a series of three resistors in the internal circuitry
- ☐ c. It has voltage levels of 5V in the internal circuitry
- ☐ d. None of these

Feedback

The correct answer is: It has a series of three resistors in the internal circuitry

Question 8

Not answered
Marked out of 1.00

Flag question

Question text

A non-inverting Op-Amp has a gain of 61 with $R_1 = 2 \text{ k}\Omega$. The resistor in the feedback path must have a value of

Select one:

- ☐ a. 62 $\text{k}\Omega$
- ☐ b. 121 $\text{k}\Omega$
- ☐ c. 61 $\text{k}\Omega$
- ☐ d. 120 $\text{k}\Omega$

Feedback

The correct answer is:
120 $\text{k}\Omega$

Question 9

Not answered

Marked out of 1.00

 Flag question

The frequency of oscillations generated by Colpitts oscillator is

Select one:

☐ a.

$$\frac{1}{2\pi\sqrt{L_{eq}C}}$$

☐ b.

$$\frac{1}{2\pi LC}$$

☐ c.

$$\frac{1}{2\pi\sqrt{LC}}$$

☐ d.

$$\frac{1}{2\pi\sqrt{LC_{eq}}}$$

$$\frac{1}{2\pi\sqrt{LC_{eq}}}$$

The correct answer is:

Question 10

Not answered

Marked out of 1.00

 Flag question

Question text

A sine wave of 0.5 V peak voltage is applied as an input to a non-inverting amplifier with $R_1 = 6 \text{ k}\Omega$ and $R_f = 24 \text{ k}\Omega$. The output voltage V_o is

Select one:

- ☐ a. 10 V peak
- ☐ b. -2.5 V peak
- ☐ c. 2.5 V peak
- ☐ d. 2.5 V peak-peak

Feedback

The correct answer is: 2.5 V peak

Question 11

Not answered
Marked out of 1.00

Flag question

Question text

The tank circuit of a Colpitt's oscillator has $L = 5 \text{ mH}$ with $C_1 = 22.22 \text{ nF}$ and $C_2 = 2.22 \text{ nF}$. The feedback factor β is

Select one:

- ☐ a. 1
- ☐ b. 0.001
- ☐ c. 0.01
- ☐ d. 0.1

Feedback

The correct answer is: 0.1

Question 12

Not answered
Marked out of 1.00

Flag question

Question text

The criterion that determines mathematical condition to generate sustained oscillations is

Select one:

- ☐ a. Shockley
- ☐ b. Threshold
- ☐ c. Pinch off
- ☐ d. Barkhausen

Feedback

The correct answer is: Barkhausen

Question 13

Not answered
Marked out of 1.00

Flag question

Question text

In an amplifier with negative feedback, open loop gain A is 20 with feedback factor as 0.04. Then the gain of the amplifier with feedback is

Select one:

- ☐ a. 11.11
- ☐ b. 200
- ☐ c. 100
- ☐ d. infinity

Feedback

The correct answer is: 11.11

Question 14

Not answered
Marked out of 1.00

Flag question

Question text

With a resistance value of $R=1\text{k}\Omega$ in a feedback network of RC oscillator frequency of oscillations generated is 5 kHz. The value of the capacitor C is..... μF

Select one:

- ☐ a. 0.0219

- ☐ b.
129
- ☐ c. 0.129
- ☐ d. 0.0129

Feedback

The correct answer is: 0.0129

Question 15

Not answered
Marked out of 1.00

Flag question

Question text

In which type of oscillator circuit capacitor split representation can be seen in tank circuit?

Select one:

- ☐ a. Weinbridge
- ☐ b. Colpitts
- ☐ c. Hartley
- ☐ d. RC phase shift

Feedback

The correct answer is: Colpitts

Question 16

Not answered
Marked out of 1.00

Flag question

Question text

In Hartley oscillator, the components used in the feedback network are

Select one:

- ☐ a. 2R and 2C
- ☐ b. 1L and 2C
- ☐ c. 2L and 2C

☐ d. 2L and 1C

Feedback

The correct answer is: 2L and 1C

Question 17

Not answered

Marked out of 1.00

Flag question

Question text

A simple application of an Op-Amp that can be used to generate a gain of unity is

Select one:

- ☐ a. Comparartor
- ☐ b. Voltage follower
- ☐ c. Differentiator
- ☐ d. Integrator

Feedback

The correct answer is: Voltage follower

Question 18

Not answered

Marked out of 1.00

Flag question

Question text

In an amplifier, positive feedback leads to

Select one:

- ☐ a. Oscillations
- ☐ b. Amplification
- ☐ c. Breakdown
- ☐ d. None of these

Feedback

The correct answer is: Oscillations

Question 19

Not answered

Marked out of 1.00

Flag question

Question text

The discharge pin in IC 555 timer is connected to ----- of the transistor.

Select one:

- ☐ a. Collector terminal
- ☐ b. Base terminal
- ☐ c. Emitter terminal
- ☐ d. None of these

Feedback

The correct answer is: Collector terminal

Question 20

Not answered
Marked out of 1.00

Flag question

Question text

Gain with negative feedback is given by $A_f = A/(1+A\beta)$. The closed loop gain is

Select one:

- ☐ a. A_f
- ☐ b. A
- ☐ c. None
- ☐ d. β

Feedback

The correct answer is: A_f

Question 21

Not answered
Marked out of 1.00

Flag question

Question text

With reference to the output voltage of an inverting Op Amp summer $V_o = -(R_f/R_1 V_1 + R_f/R_2 V_2 + R_f/R_3 V_3)$

Select one:

- ☐ a. $V_o = -(V_1 + V_2 + V_3)$
- ☐ b. $V_o = -V_1 + V_2 + V_3$
- ☐ c. $V_o = V_1 + V_2 + V_3$

Feedback

The correct answer is: $V_o = -(V_1 + V_2 + V_3)$

Question 22

Not answered

Marked out of 1.00

Flag question

The output voltage of an Op Amp differentiator is given by

Select one:

- ☐ a. $V_o = -R_f C \frac{dV_{in}}{dt}$
- ☐ b. $V_o = -\frac{1}{C_f} \int_0^t V_{in} dt$
- ☐ c. $V_o = -\frac{1}{R_f} \int_0^t V_{in} dt$
- ☐ d. $V_o = -\frac{1}{R_f C} \frac{dV_{in}}{dt}$

The correct answer is: $V_o = -R_f C \frac{dV_{in}}{dt}$

Question 23

Not answered

Marked out of 1.00

Flag question

Question text

An inverting Op-Amp has a gain of - 61 with $R_1 = 1 \text{ k}\Omega$. The resistor in the feedback path must have a value of

Select one:

- ☐ a. 62 k Ω
- ☐ b. 162k Ω
- ☐ c. 82 k Ω
- ☐ d. 182 k Ω

Feedback

The correct answer is: 162k Ω

Question 24

Not answered

Marked out of 1.00

Flag question

Question text

Which is not the ideal characteristic of an op-amp?

Select one:

- ☐ a. Input Resistance is zero
- ☐ b. Output impedance is zero
- ☐ c. Bandwidth is infinity
- ☐ d. Open loop voltage gain is infinity

Feedback

The correct answer is: Input Resistance is zero

Question 25

Not answered

Marked out of 1.00

Flag question

Question text

The components in the feedback network of LC oscillators are

Select one:

- ☐ a. Only L Component
- ☐ b. L & C components
- ☐ c. R & C components

☐ d. None of these

Feedback

The correct answer is: L & C components

Question 26

Not answered

Marked out of 1.00

Flag question

Question text

In an RC phase shift oscillator, the resistances in the feedback network are $4.7\text{ k}\Omega$ and capacitor values are $C = 0.47\mu\text{F}$. The frequency of oscillations is

Select one:

- ☐ a. 294.13 Hz
- ☐ b. 29.413 Hz
- ☐ c. 2941.3 Hz
- ☐ d. None of these

Feedback

The correct answer is: 29.413 Hz

Question 27

Not answered

Marked out of 1.00

Flag question

Question text

RC phase shift oscillator is a

Select one:

- ☐ a. High frequency oscillator
- ☐ b. Low frequency oscillator
- ☐ c. Relaxation oscillator
- ☐ d. Stable frequency oscillator

Feedback

The correct answer is: Low frequency oscillator

Question 28

Not answered

Marked out of 1.00

Flag question

Question text

In a practical oscillator circuit, to start oscillations, the loop gain $A\beta$ must be

Select one:

- ☐ a. Not equal to 1
- ☐ b. Greater than 1
- ☐ c. Equal to 1
- ☐ d. Less than 1

Feedback

The correct answer is: Greater than 1

Question 29

Not answered

Marked out of 1.00

Flag question

Question text

The duty cycle 'D' of IC 555 timer in astable mode in terms of its time period is defined by

Select one:

- ☐ a. T/T_{off}
- ☐ b. $T_{on}/[T_{on}+T_{off}]$
- ☐ c. T/T_{on}
- ☐ d. T_{on}/T_{off}

Feedback

The correct answer is: $T_{on}/[T_{on}+T_{off}]$

Question 30

Not answered

Marked out of 1.00

 Flag question

For an amplifier with positive feedback, the closed loop gain is given by

Select one:

☐ a. $A_f = \frac{A}{1-A\beta}$

☐ b. $A_f = \frac{A}{1+A\beta}$

☐ c. $A_f = \frac{A}{1-\beta}$

☐ d. $A_f = \frac{A}{1-v\beta}$

Question 31

Not answered

Marked out of 1.00

 Flag question

Question text

Gain with negative feedback is given by $A/1-A\beta$. The feedback factor is

Select one:

- ☐ a. A_f
- ☐ b. β
- ☐ c. A
- ☐ d. None

Feedback

The correct answer is: β

Question 32

Not answered

Marked out of 1.00

 Flag question

Question text

An RC network in an RC phase shift oscillator has $C = 0.2 \mu\text{F}$. The frequency of oscillations is 1 kHz. The value of R used in the feedback network is

Select one:

- ☐ a. 600Ω
- ☐ b. 500Ω
- ☐ c. $1\text{K}\Omega$
- ☐ d. 325Ω

Feedback

The correct answer is:
 325Ω

Question 33

Not answered
Marked out of 1.00

Flag question

Question text

Negative feedback amplifier

Select one:

- ☐ a. Increases gain
- ☐ b. Increases noise
- ☐ c. Reduces gain
- ☐ d. Reduces phase shift

Feedback

The correct answer is: Reduces gain

Question 34

Not answered
Marked out of 1.00

Flag question

Question text

An inverting Op-Amp has a gain of - 61 with $R_1 = 1 \text{ k}\Omega$. The resistor in the feedback path must have a value of

Select one:

- ☐ a. 61 k Ω
- ☐ b. 60 k Ω
- ☐ c. 62 k Ω
- ☐ d. 59 k Ω

Feedback

The correct answer is: 61 k Ω

Question 35

Not answered
Marked out of 1.00

Flag question

Question text

An amplifier has an open loop voltage gain of 1000. If 10% negative voltage series feedback is used, then the closed loop gain is

Select one:

- ☐ a. 990
- ☐ b. 9.9
- ☐ c. 0.9
- ☐ d. 99.9

Feedback

The correct answer is: 9.9

Question 36

Not answered
Marked out of 1.00

Flag question

Question text

Hartley oscillator is a

Select one:

- ☐ a. High frequency oscillator
- ☐ b. Relaxation oscillator
- ☐ c. Stable frequency oscillator

☐ d. Low frequency oscillator

Feedback

The correct answer is: High frequency oscillator

Question 37

Not answered

Marked out of 1.00

Flag question

Question text

The voltage levels fixed at one of the terminals of two comparators in the internal circuitry of IC 555 timer are

Select one:

- ☐ a. V_{cc} and $(2/5)V_{cc}$
- ☐ b. $(1/5)V_{cc}$ and $(2/5)V_{cc}$
- ☐ c. $(1/3)V_{cc}$ and $(2/3)V_{cc}$
- ☐ d. 0V and $(2/5)V_{cc}$

Feedback

The correct answer is: $(1/3)V_{cc}$ and $(2/3)V_{cc}$

Question 38

Not answered

Marked out of 1.00

Flag question

Question text

An amplifier has an open loop voltage gain of 2000. If 40% negative voltage series feedback is used, then the closed loop gain is

Select one:

- ☐ a. 0.249
- ☐ b. 2.49
- ☐ c. 249
- ☐ d. 24.9

Feedback

The correct answer is: 2.49

Question 39

Not answered

Marked out of 1.00

Flag question

Question text

An integrator circuit using an Op Amp hasin its feedback path

Select one:

- ☐ a. Inductor
- ☐ b. Capacitor
- ☐ c. Diode
- ☐ d. Resistor

Feedback

The correct answer is: Capacitor

Question 40

Not answered

Marked out of 1.00

Flag question

Question text

An amplifier has an open loop voltage gain of 100,000. If the negative voltage series feedback factor is 0.01, then the closed loop gain is

Select one:

- ☐ a. 9.9
- ☐ b. 990
- ☐ c. 0.9
- ☐ d. 99.9

Feedback

The correct answer is: 99.9

Question 41

Not answered

Marked out of 1.00

Flag question

Question text

A differentiator circuit using an Op Amp hasin its feedback path

Select one:

- ☐ a. Inductor
- ☐ b. Capacitor
- ☐ c. Diode
- ☐ d. Resistor

Feedback

The correct answer is: Resistor

Question 42

Not answered

Marked out of 1.00

Flag question

Question text

IC 555 timer working as a free running oscillator is a

Select one:

- ☐ a. DC to DC inverter
- ☐ b. DC to AC converter
- ☐ c. AC to DC converter
- ☐ d. DC to DC converter

Feedback

The correct answer is: DC to AC converter

Question 43

Not answered

Marked out of 1.00

Flag question

Question text

Op Amp Comparator is a circuit whose output voltage switches between

Select one:

- ☐ a. $+V_{in}, -V_{in}$
- ☐ b. $+V_{sat}, -V_{sat}$
- ☐ c. None
- ☐ d. $+V_o, -V_o$

Feedback

The correct answer is: $+V_{sat}, -V_{sat}$

Question 44

Not answered

Marked out of 1.00

Flag question

Question text

In an RC-Phase shift oscillator, the components used in the feedback network are

Select one:

- ☐ a. $3R$ and $3C$
- ☐ b. $2R$ and $2L$
- ☐ c. $1R$ and $1C$
- ☐ d. $4R$ and $4C$

Feedback

The correct answer is: $3R$ and $3C$

Question 45

Not answered

Marked out of 1.00

Flag question

Question text

In Colpitts' oscillator, the components used in the feedback network are

Select one:

- ☐ a. $2L$ and $1C$
- ☐ b. $2R$ and $2C$

- ☐ c. 2C and 1L
- ☐ d. 2L and 2C

Feedback

The correct answer is: 2C and 1L

Question 46

Not answered

Marked out of 1.00

Flag question

Question text

In a Hartley oscillator, if $L_1 = 7 \text{ mH}$, $L_2 = 10 \text{ mH}$ and $C = 0.01 \mu\text{F}$, the value of the feedback factor β is

Select one:

- ☐ a.
- 0.7
- ☐ b.
- 70
- ☐ c. 700
- ☐ d.
- 0.07

Feedback

The correct answer is:

0.7

Question 47

Not answered

Marked out of 1.00

 Flag question

The frequency of oscillations generated by Hartley oscillator is

Select one:

☐ a.

$$\frac{1}{2\pi LC}$$

☐ b.

$$\frac{1}{2\pi\sqrt{LC_{eq}}}$$

☐ c.

$$\frac{1}{2\pi\sqrt{L_{eq}C}}$$

☐ d.

$$\frac{1}{2\pi\sqrt{LC}}$$

The correct answer is:

$$\frac{1}{2\pi\sqrt{L_{eq}C}}$$

Question 48

Not answered

Marked out of 1.00

 Flag question

Question text

For an Op-Amp having differential gain A_d and Common mode gain A_c , CMRR is

Select one:

☐ a. $1 + (A_d/A_c)$ ☐ b. A_c/A_d



c.

A_d/A_c



d. $A_d + A_c$

Feedback

The correct answer is:

A_d/A_c

Question 49

Not answered

Marked out of 1.00

Flag question

Question text

A non-inverting closed loop op amp circuit generally has a gain factor

Select one:



a. Equal to one



b. Less than one



c. Zero



d. Greater than one

Feedback

The correct answer is: Greater than one

Question 50

Not answered

Marked out of 1.00

Flag question

Question text

A sine wave of 0.5 V peak voltage is applied as an input to a non-inverting amplifier with $R_1 = 10\text{ k}\Omega$ and $R_f = 50\text{ k}\Omega$ The output voltage V_o is

Select one:



a. -3 V peak



b. 3 V peak-peak



c. 3 V peak

☐ d. 2.5 V peak

Feedback

The correct answer is: 3 V peak

Question 51

Not answered

Marked out of 1.00

Flag question

Question text

The duty cycle D in astable mode using IC 555 is obtained using

Select one:

- ☐ a. RS Flip flop
- ☐ b. Internally connected resistors
- ☐ c. External resistors
- ☐ d. Comparators

Feedback

The correct answer is: External resistors

Question 52

Not answered

Marked out of 1.00

Flag question

Question text

Which among the following parameters acts as an initiator for the operation of an oscillator in the absence of input signal?

Select one:

- ☐ a. Noise current
- ☐ b. Noise power
- ☐ c. Noise temperature
- ☐ d. Noise voltage

Feedback

The correct answer is: Noise voltage

Question 53

Not answered
Marked out of 1.00

Flag question

Question text

In which type of oscillator circuit inductor split representation can be seen in tank circuit?

Select one:

- ☐ a. Wein bridge
- ☐ b. Hartley
- ☐ c. RC phase shift
- ☐ d. Colpitts

Feedback

The correct answer is: Hartley

Question 54

Not answered
Marked out of 1.00

Flag question

Question text

An inverting amplifier using Op Amp with as the resistors provide an output voltage equal to one of them as

Select one:

- ☐ a. $V_o = [1+(R_f/R_1)] V_{in}$
- ☐ b. $V_o = (R_f/R_1) V_{in}$
- ☐ c. $V_o = [1-(R_f/R_1)] V_{in}$
- ☐ d. $V_o = - (R_f/R_1) V_{in}$

Feedback

The correct answer is: $V_o = - (R_f/R_1) V_{in}$

Question 55

Not answered
Marked out of 1.00

Flag question

Question text

In a Hartley oscillator, if $L_1 = 5 \text{ mH}$, $L_2 = 10 \text{ mH}$ and $C = 0.01 \mu\text{F}$, the value of the feedback factor β is

Select one:

- ☐ a. 50
- ☐ b. 500
- ☐ c. 0.5
- ☐ d. 0.05

Feedback

The correct answer is:
0.5

Question 56

Not answered
Marked out of 1.00

Flag question

Question text

A voltage follower using Op Amp is also known as

Select one:

- ☐ a. Switch
- ☐ b. Non inverting amplifier
- ☐ c. Buffer amplifier
- ☐ d. Inverting amplifier

Feedback

The correct answer is: Buffer amplifier

Question 57

Not answered
Marked out of 1.00

Flag question

Question text

In a Colpitts oscillator, if $C_1 = 100 \text{ pF}$, $C_2 = 60 \text{ pF}$ and $L = 0.422 \text{ H}$, the value of the feedback factor β is

Select one:

- ☐ a. 600
- ☐ b. 0.06
- ☐ c. 60
- ☐ d. 0.6

Feedback

The correct answer is: 0.6

Question 58

Not answered
Marked out of 1.00

Flag question

Question text

A non-inverting Op-Amp has a gain of 61 with $R_1 = 1 \text{ k}\Omega$. The resistor in the feedback path must have a value of

Select one:

- ☐ a. 60 k Ω
- ☐ b. 62 k Ω
- ☐ c. 61 k Ω
- ☐ d. 59 k Ω

Feedback

The correct answer is: 60 k Ω

Question 59

Not answered

Marked out of 1.00

Flag question

The charging time or ON time of the capacitor in Astable mode of operation using IC 555 timer is

Select one:

- ☐ a. $T_{ON} = 0.693(R_1)C$
- ☐ b. $T_{ON} = 0.693(R_2)C$
- ☐ c. $T_{ON} = 0.693(F)C$
- ☐ d. $T_{ON} = 0.693(R_1 + R_2)C$

The correct answer is: $T_{ON} = 0.693(R_1 + R_2)C$

Question 60

Not answered

Marked out of 1.00

Flag question

Question text

The frequency of oscillations generated by RC phase shift oscillator is

Select one:

☐ a. $12\pi 6RC12\sqrt{6}$

☐ b. $12\pi RC12\sqrt{6}$

☐ c. $12\pi 6\sqrt{RC12\sqrt{6}}$

☐ d. $12\pi 6RC\sqrt{12\sqrt{6}}$

Feedback

The correct answer is:

$12\pi 6\sqrt{RC12\sqrt{6}}$

Question 61

Not answered

Marked out of 1.00

[Flag question](#)

Question text

Which of the following electrical characteristics is not exhibited by an ideal op-amp?

Select one:

- ☐ a. Infinite output resistance
- ☐ b. Infinite bandwidth
- ☐ c. Infinite slew rate
- ☐ d. Infinite voltage gain

Feedback

The correct answer is: Infinite output resistance

Question 62

Not answered

Marked out of 1.00

[Flag question](#)

Question text

The feedback factor of Hartley oscillator is given by

Select one:

- ☐ a. $\beta = A_2/A_1$
- ☐ b. $\beta = L_1/L_2$
- ☐ c. $\beta = A_1/A_2$
- ☐ d. $\beta = L_2/L_1$

Feedback

The correct answer is: $\beta = L_1/L_2$

Question 63

Not answered

Marked out of 1.00

Flag question

A non-inverting amplifier using Op Amp with R_1 and R_f as the resistors provide an output voltage equal to one of them as

Select one:

- ☐ a. $V_0 = \left(1 + \frac{R_f}{R_1}\right) V_{in}$
- ☐ b. $V_0 = \left(\frac{R_f}{R_1}\right) V_{in}$
- ☐ c. $V_0 = -\left(1 + \frac{R_f}{R_1}\right) V_{in}$
- ☐ d. $V_0 = -\left(\frac{R_f}{R_1}\right) V_{in}$

The correct answer is: $V_0 = \left(1 + \frac{R_f}{R_1}\right) V_{in}$

Question 64

Not answered

Marked out of 1.00

Flag question

Question text

For an ideal op-amp, which of the following is false?

Select one:

- ☐ a. The current from output terminal is zero
- ☐ b. The output resistance is zero
- ☐ c. The differential voltage between the input terminals is zero
- ☐ d. The current into the input terminals is zero

Feedback

The correct answer is: The current from output terminal is zero

Question 65

Not answered

Marked out of 1.00

Flag question

The duty cycle 'D' of IC 555 timer in astable mode in terms of resistors used in the external circuit is defined by

Select one:

- ☐ a. $\frac{R_1 + 2R_2}{R_1 + R_2}$
- ☐ b. $\frac{R_1 + R_2}{R_1 - 2R_2}$
- ☐ c. $\frac{R_1 - R_2}{R_1 - 2R_2}$
- ☐ d. $\frac{R_1 + R_2}{R_1 + 2R_2}$

The correct answer is: $\frac{R_1 + R_2}{R_1 + 2R_2}$

Question 66

Not answered

Marked out of 1.00

Flag question

The sustained oscillations in an oscillator circuit is obtained by setting the loop gain $A\beta$ as

Select one:

- ☐ a. Less than 1
- ☐ b. Equal to 1
- ☐ c. Not equal to 1
- ☐ d. Greater than 1

The correct answer is: Greater than 1

Question 67

Not answered

Marked out of 1.00

Flag question

Question text

In a Hartley oscillator, if $L_1 = 2 \text{ mH}$, $L_2 = 8 \text{ mH}$ and $C = 0.01 \mu\text{F}$, the value of the feedback factor β is

Select one:

- ☐ a. 25
- ☐ b. 250
- ☐ c. 0.025
- ☐ d. 0.25

Feedback

The correct answer is: 0.25

Question 68

Not answered

Marked out of 1.00

Flag question

Question text

The pin number 7 in IC 555 timer is assigned as

Select one:

- ☐ a. None of these
- ☐ b. Discharge pin
- ☐ c. Threshold pin
- ☐ d. output pin

Feedback

The correct answer is: Discharge pin

Question 69

Not answered

Marked out of 1.00

Flag question

Question text

What is an angle of phase shift for each designed RC network in the Phase Shift Oscillator circuit?

Select one:

- ☐ a. 60°
- ☐ b. 30°
- ☐ c. 90°
- ☐ d. 45°

Feedback

The correct answer is: 60°

Question 70

Not answered

Marked out of 1.00

Flag question

Question text

The output voltage obtained for an ideal op-amp is by

Select one:

- ☐ a. Amplifying the difference between the two input voltages
- ☐ b. Amplifying products of two input voltage
- ☐ c. Amplifying individual input voltages
- ☐ d. None of the mentioned

Feedback

The correct answer is: Amplifying the difference between the two input voltages

UNIT-3

Question 1

Complete

Mark 0.00 out of 1.00

Flag question

Question text

Which of the following is true about microprocessors?

Select one:

- ☐ a. It contains ALU, CU, and registers
- ☐ b. It has interfacing circuits
- ☐ c. It uses Harvard architecture
- ☒ d. It has an internal memory

Feedback

The correct answer is: It contains ALU, CU, and registers

Question 2

Complete

Mark 0.00 out of 1.00

Flag question

Question text

The process of transferring a mobile station from one base station to another base station is called

Select one:

- ☐ a. a) Channel efficiency
- ☒ b. Frequency Division Duplexing
- ☐ c. Hand-off
- ☐ d. Frequency reuse

Feedback

The correct answer is: Hand-off

Question 3

Complete

Mark 0.00 out of 1.00

Flag question

Question text

A ____ provides fast, discrete-time, signal-processing instructions

Select one:

- ☐ a. GPP
- ☐ b. ASSP
- ☒ c. DPS
- ☐ d. DSP

Feedback

The correct answer is: DSP

Question 4

Not answered

Marked out of 1.00

Flag question

Question text

The unit used for measuring Message or information is

Select one:

- ☐ a. meter per second
- ☐ b. Ohms
- ☐ c. Hertz
- ☐ d. bits per second

Feedback

The correct answer is: bits per second

Question 5

Not answered
Marked out of 1.00

Flag question

Question text

If there are 4 subgroups within a frequency band, for a cellular system consisting of a total of 7 cells, the total number of available radio channels is _____.

Select one:

- ☐ a. 3
- ☐ b. 56
- ☐ c. 11
- ☐ d. 28

Feedback

The correct answer is: 28

Question 6

Not answered
Marked out of 1.00

Flag question

Question text

The following frequency band is utilized in cellular or mobile communication system

Select one:

- ☐ a. Audible frequency
- ☐ b. Audio frequency
- ☐ c. Radio frequency
- ☐ d. Unlicensed frequency

Feedback

The correct answer is: Radio frequency

Question 7

Not answered
Marked out of 1.00

Flag question

Question text

In a cellular radio system, the _____ connects the cellular base stations and mobile stations to the Public Switched Telephone Networks.

Select one:

- ☐ a. Mobile Switching Center (MSO)
- ☐ b. Time Division Multiple Access(TDMA)
- ☐ c. Forward Control Channel (FCC)
- ☐ d.

Base station(BS)

Feedback

The correct answer is:
Mobile Switching Center (MSO)

Question 8

Not answered
Marked out of 1.00

Flag question

Question text

Which of the following channels used for communication has highest bandwidth?

Select one:

- ☐ a. Fibre optic cable
- ☐ b. Co axial cable
- ☐ c. Flat cable
- ☐ d. Pair of wires

Feedback

The correct answer is: Fibre optic cable

Question 9

Not answered
Marked out of 1.00

Flag question

Question text

The initial mobile communication systems for public safety in United states used the following modulation technique,

Select one:

- ☐ a. Phase modulation
- ☐ b. Time based modulation
- ☐ c. Frequency modulation
- ☐ d. Amplitude modulation

Feedback

The correct answer is:
Amplitude modulation

Question 10

Not answered
Marked out of 1.00

Flag question

Question text

The speech signal frequency range is

Select one:

- ☐ a. 300 Hz to 3.4 kHz
- ☐ b. 10 to 10 kHz
- ☐ c. 30 to 34 kHz
- ☐ d. 20 Hz to 20 kHz

Feedback

The correct answer is: 300 Hz to 3.4 kHz

Question 11

Not answered

Marked out of 1.00

Flag question

Question text

The mechanism of using the same frequency band within a geographical area in a Cellular or mobile communication system is referred to as,

Select one:

- ☐ a. Reliability
- ☐ b. Bandwidth coordination
- ☐ c. Efficiency
- ☐ d. Frequency reuse

Feedback

The correct answer is: Frequency reuse

Question 12

Not answered

Marked out of 1.00

Flag question

Question text

The loss in the transmission is minimum for the following type of wire

Select one:

- ☐ a. Flat cable
- ☐ b. Fiber optic cable
- ☐ c. Coaxial cable
- ☐ d. Copper wire

Feedback

The correct answer is: Fiber optic cable

Question 13

Not answered

Marked out of 1.00

Flag question

Question text

Best example for man-made noise is

Select one:

- ☐ a. Radiations from sun, stars
- ☐ b. Rain
- ☐ c. Traffic noise
- ☐ d. Reflections from earth

Feedback

The correct answer is: Traffic noise

Question 14

Not answered

Marked out of 1.00

Flag question

Question text

In a cellular system, the channel used for voice transmission from the base station to the mobile station is called _____ .

Select one:

- ☐ a. Forward Voice Channel (FVC)
- ☐ b. Reverse Control Channel(RCC)

- ☐ c. Forward Control Channel(FCC)
- ☐ d. Reverse Voice Channel (RVC)

Feedback

The correct answer is: Forward Voice Channel (FVC)

Question 15

Not answered
Marked out of 1.00

Flag question

Question text

The inherent interference resistance property between wireless cellular channels is observed in

Select one:

- ☐ a. Time Division Multiple Access
- ☐ b. Frequency Division Multiple Access
- ☐ c. Code Division Multiple Access
- ☐ d. Space Division Multiple Access

Feedback

The correct answer is: Code Division Multiple Access

Question 16

Not answered
Marked out of 1.00

Flag question

Question text

Which of the following is true about Embedded System?

Select one:

- ☐ a. All of them
- ☐ b. An embedded system usually performs a specialized operation and does the same repeatedly
- ☐ c. It must have connected peripherals to connect input and output devices
- ☐ d. It must have a memory, as its software usually embeds in ROM

Feedback

The correct answer is: All of them

Question 17

Not answered

Marked out of 1.00

Flag question

Question text

Which of the following statements are true for von Neumann architecture?

Select one:

- ☐ a. External bus for data memory only
- ☐ b. Shared bus between the program memory and data memory
- ☐ c. Separate bus between the program memory and data memory
- ☐ d. External bus for program memory and data memory

Feedback

The correct answer is: Shared bus between the program memory and data memory

Question 18

Not answered

Marked out of 1.00

Flag question

Question text

Princeton architecture is also known as

Select one:

- ☐ a. CISC
- ☐ b. Von Neumann architecture
- ☐ c. Harvard
- ☐ d. RISC

Feedback

The correct answer is: Von Neumann architecture

Question 19

Not answered

Marked out of 1.00

Flag question

Question text

The function of a sensor is to _____

Select one:

- ☐ a. None
- ☐ b. Separate physical parameters
- ☐ c. Only Track the data
- ☐ d. Detect events within specified environment

Feedback

The correct answer is: Detect events within specified environment

Question 20

Not answered

Marked out of 1.00

Flag question

Question text

ASIC stands for?

Select one:

- ☐ a. Avionics Subsystem Interface Contractor
- ☐ b. Application Specific Integrated Circuit
- ☐ c. Aviation Security Identification Card
- ☐ d. Application-Specific introduction code

Feedback

The correct answer is: Application Specific Integrated Circuit

Question 21

Not answered

Marked out of 1.00

Flag question

Question text

Which of the following is example for the input subsystem of an embedded system?

Select one:

- ☐ a. None of these
- ☐ b. Seven Segment

- ☐ c. LED
- ☐ d. Optocoupler

Feedback

The correct answer is: Optocoupler

Question 22

Not answered
Marked out of 1.00

Flag question

Question text

RISC stands for

Select one:

- ☐ a. Remaining Intermediate Storage of Computer
- ☐ b. Reduced Intermediate Storage of Computer
- ☐ c. Reduced instruction set computer
- ☐ d. Remaining Instruction Set of Computer

Feedback

The correct answer is: Reduced instruction set computer

Question 23

Not answered
Marked out of 1.00

Flag question

Question text

Which of the following is true about optocouplers?

Select one:

- ☐ a. None of these
- ☐ b. Optocoupler acts as an input device only
- ☐ c. Optocoupler acts as an output device only
- ☐ d. Optocoupler can be used in both input and output circuitry

Feedback

The correct answer is: Optocoupler can be used in both input and output circuitry

Question 24

Not answered

Marked out of 1.00

Flag question

Question text

Which one of the following offers external chips for memory and peripheral interface circuits?

Select one:

- ☐ a. Embedded system
- ☐ b. Microcontroller
- ☐ c. Peripheral System
- ☐ d. Microprocessor

Feedback

The correct answer is: Microprocessor

Question 25

Not answered

Marked out of 1.00

Flag question

Question text

----- converts the information into signal suitable to be transmitted over the communication medium.

Select one:

- ☐ a. Receiver
- ☐ b. Transmitter
- ☐ c. Transmission line
- ☐ d. Channel

Feedback

The correct answer is: Transmitter

Question 26

Not answered

Marked out of 1.00

Flag question

Question text

The radio channel used for transmission of information from a base station to a mobile station is called,

Select one:

- ☐ a. Control channel
- ☐ b. Reverse channel
- ☐ c. Forward channel
- ☐ d. Mobile channel

Feedback

The correct answer is: Forward channel

Question 27

Not answered

Marked out of 1.00

Flag question

Question text

Von-Neumann Architecture is Simple in design.

Select one:

- ☐ True
- ☐ False

Feedback

The correct answer is 'True'.

Question 28

Not answered

Marked out of 1.00

Flag question

Question text

A computer that uses the same memory space for both data and program instructions is classified as ____.

Select one:

- ☐ a. Harvard architecture
- ☐ b. Von Neumann architecture
- ☐ c. None of the these
- ☐ d. Memory architecture

Feedback

The correct answer is: Von Neumann architecture

Question 29

Not answered
Marked out of 1.00

Flag question

Question text

Harvard architecture has _____

Select one:

- ☐ a. Dedicated buses for data and program memory
- ☐ b. Pipeline technique
- ☐ c. all of the mentioned
- ☐ d. Complex architecture

Feedback

The correct answer is: all of the mentioned

Question 30

Not answered
Marked out of 1.00

Flag question

Question text

The audio frequency range is

Select one:

- ☐ a. 10 to 10 kHz
- ☐ b. 20 Hz to 20 kHz
- ☐ c. 30 to 34 kHz
- ☐ d. 300 Hz to 3.4 kHz

Feedback

The correct answer is: 20 Hz to 20 kHz

Question 31

Not answered
Marked out of 1.00

Flag question

Question text

Fibre optic cable is a ----- type of channel.

Select one:

- ☐ a. Wired channel
- ☐ b. Wireless channel
- ☐ c. Free space channel
- ☐ d. Radio channel

Feedback

The correct answer is: Wired channel

Question 32

Not answered
Marked out of 1.00

Flag question

Question text

CISC stands for

Select one:

- ☐ a. Complex Instruction Set Computer
- ☐ b. Complete Instruction Set Compliment
- ☐ c. Computer Instruction Set Compliment
- ☐ d. Computer Indexed Set Components

Feedback

The correct answer is: Complex Instruction Set Computer

Question 33

Not answered

Marked out of 1.00

Flag question

Question text

Actuators are used to

Select one:

- ☐ a. Sense an object
- ☐ b. Make a mechanical movement
- ☐ c. All of the above
- ☐ d. Activate a chemical

Feedback

The correct answer is: Make a mechanical movement

Question 34

Not answered

Marked out of 1.00

Flag question

Question text

Processor which is complex and expensive to produce is _____

Select one:

- ☐ a. RISC
- ☐ b. CISC
- ☐ c. Multi-core
- ☐ d. EPIC

Feedback

The correct answer is: CISC

Question 35

Not answered
Marked out of 1.00

Flag question

Question text

A microcontroller at-least should consist of:

Select one:

- ☐ a. RAM, ROM, I/O ports and timers
- ☐ b. CPU, RAM, ROM, I/O ports and timers
- ☐ c. CPU, ROM, I/O ports and timers
- ☐ d. CPU, RAM, I/O ports and timers

Feedback

The correct answer is: CPU, RAM, ROM, I/O ports and timers

Question 36

Not answered
Marked out of 1.00

Flag question

Question text

Best example for natural noise is

Select one:

- ☐ a. Industrial noise
- ☐ b. Traffic noise
- ☐ c. Rain



d. Sound pollution

Feedback

The correct answer is: Rain
